Planetesimal Formation under Realistic Gas Dynamics

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Hydrodynamics of Circumstellar Disks and Planet Formation, PFITS+
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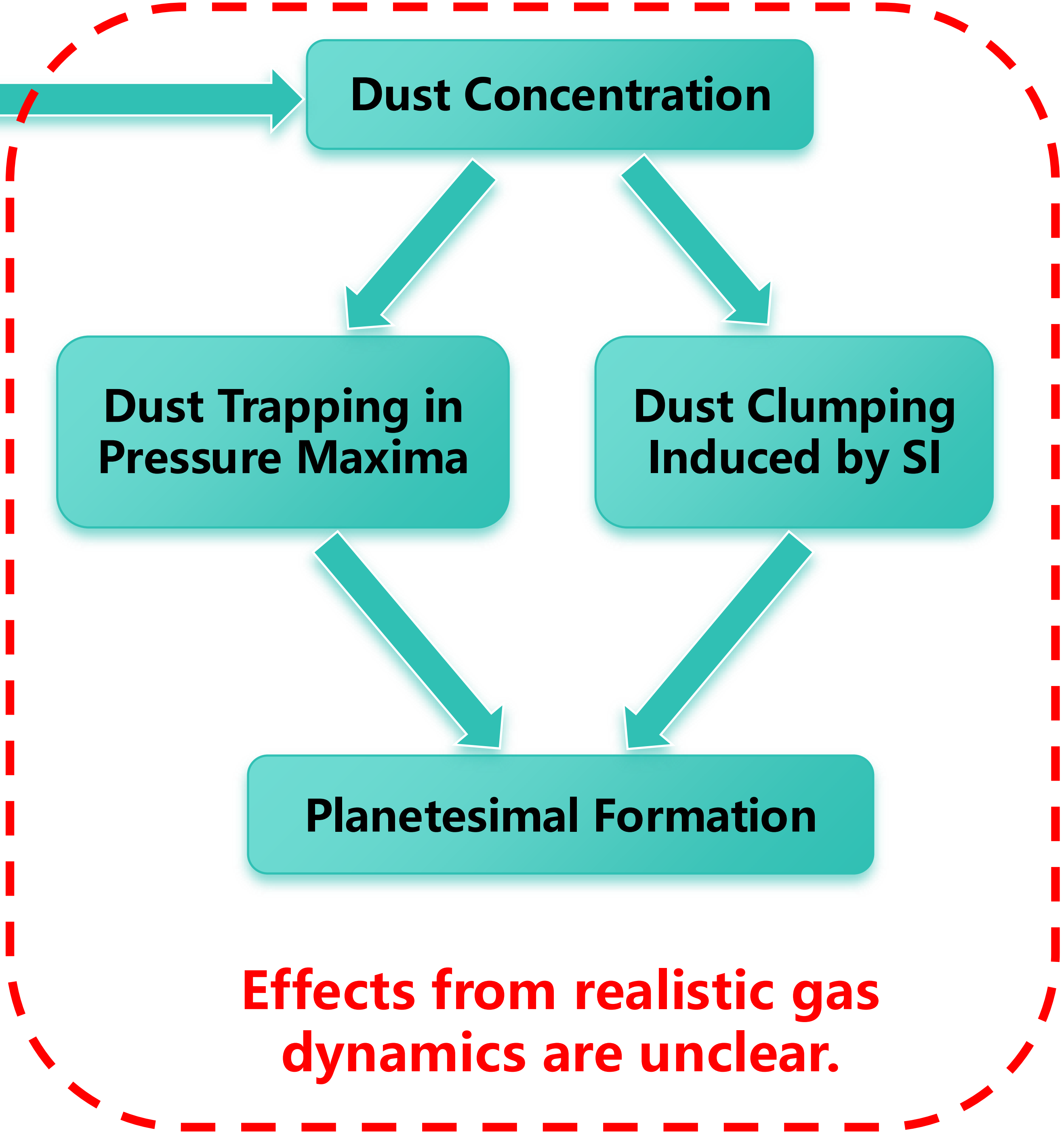
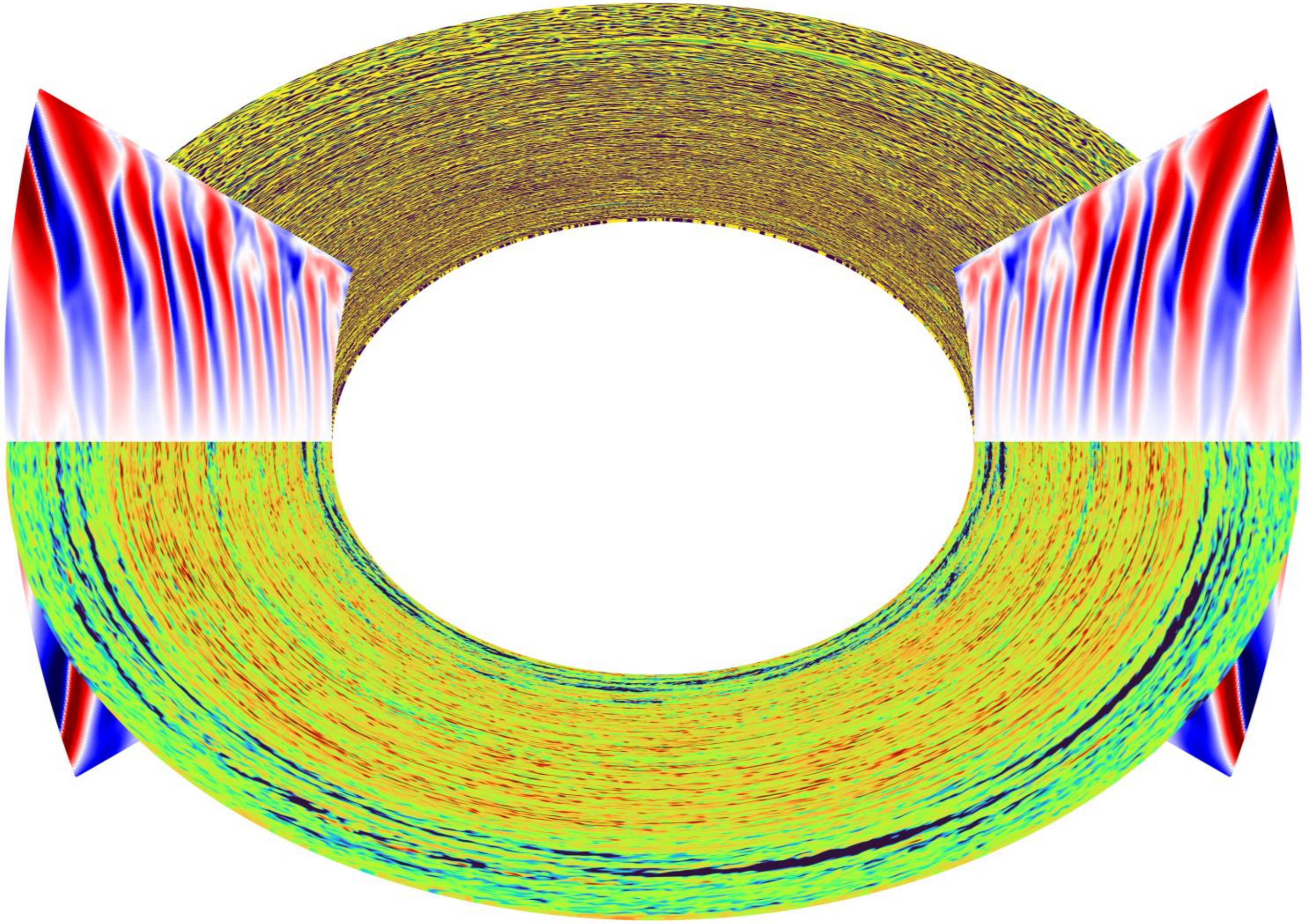
Bottom-Up Picture of Planet Formation



Based on **first-principles modeling**, we investigate dust concentration in a self-consistent turbulent and wind-launching environment.

Planetesimal Formation under Realistic Gas Dynamics

I. Dust Clumping in 3D Turbulence Induced by Four Instabilities



Huang & Bai,
ApJ 2025a, ApJL 2025b

Vertical Shear Instability (VSI)

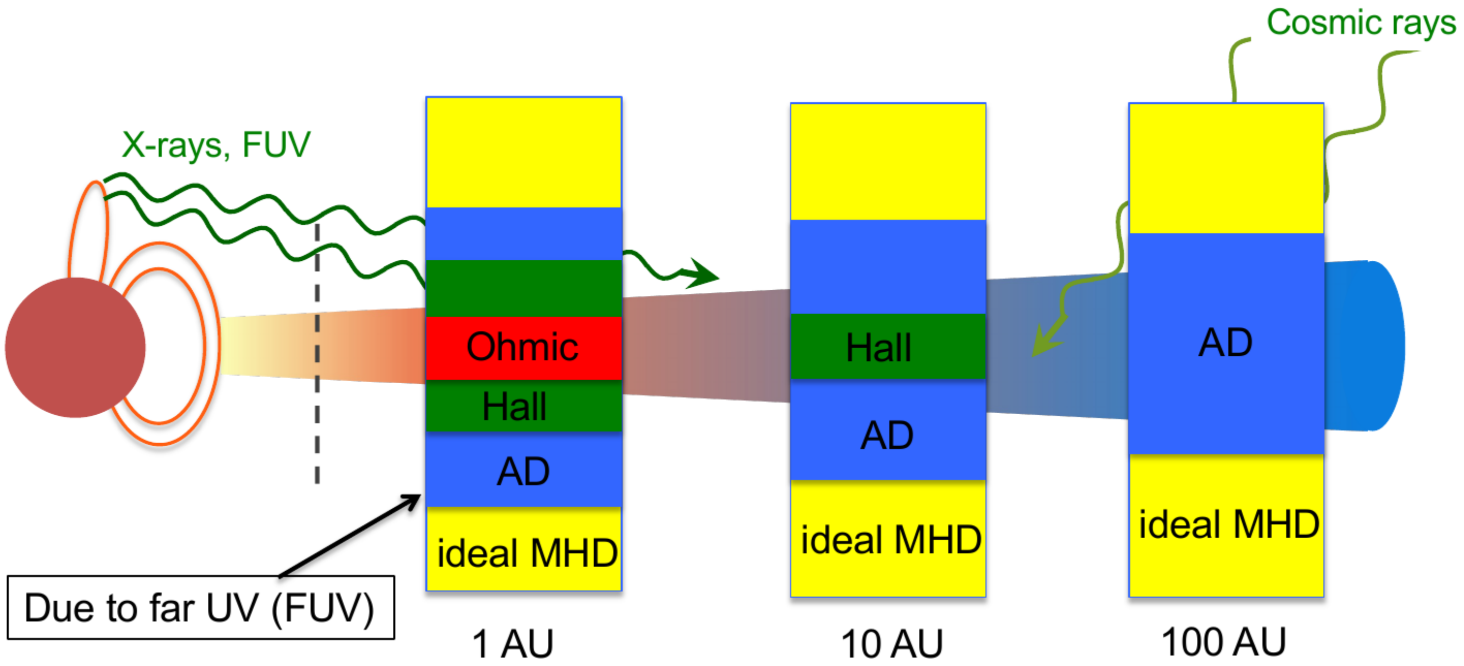
Streaming Instability (SI)

Rossby Wave Instability (RWI)

Kelvin-Helmholtz Instability (KHI)

Turbulence is not a necessarily diffusive effect that inhibits dust concentration and planetesimal formation.

II. Dust Concentration in Turbulent and Windy Protoplanetary Disks



$$\frac{\partial \boldsymbol{B}}{\partial t} = \nabla\times(\boldsymbol{v}\times\boldsymbol{B}) - \nabla\times\left[\frac{4\pi\eta}{c}\boldsymbol{J} + \frac{\boldsymbol{J}\times\boldsymbol{B}}{en_e} - \frac{(\boldsymbol{J}\times\boldsymbol{B})\times\boldsymbol{B}}{c\gamma\rho\rho_i}\right]$$

inductive **Ohmic** **Hall** **AD**

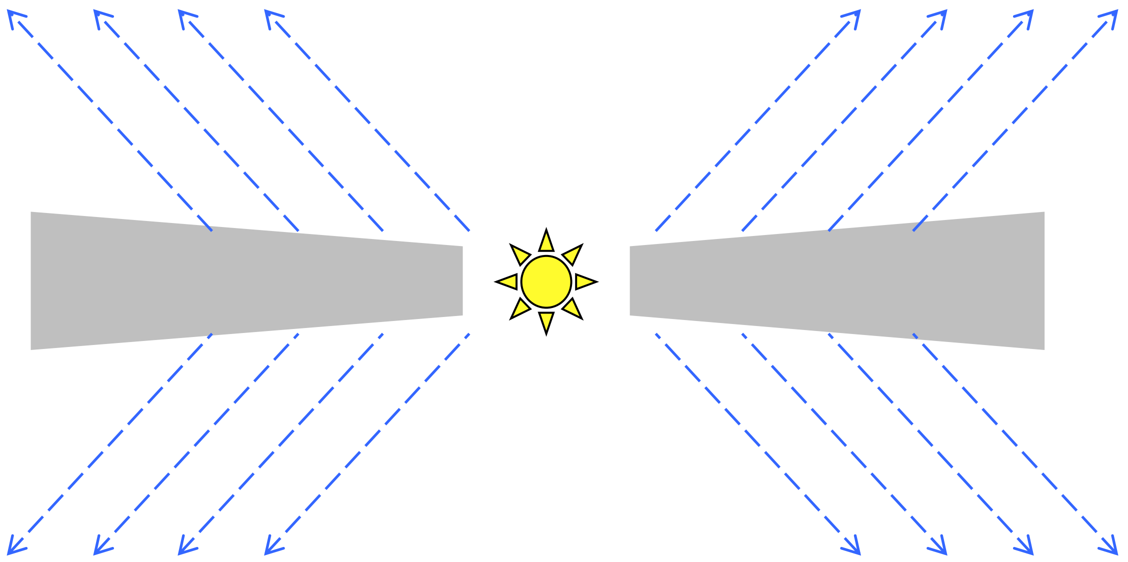
$\boxed{\sim \frac{n}{n_e}}$ $\boxed{\sim \frac{n}{n_e}\frac{B}{\rho}}$ $\boxed{\sim \frac{n}{n_e}\frac{B^2}{\rho^2}}$

We carry 2D axisymmetric **MHD + dust + AD** simulations using the multifluid dust module of Athena++ (Huang & Bai, to be submitted).

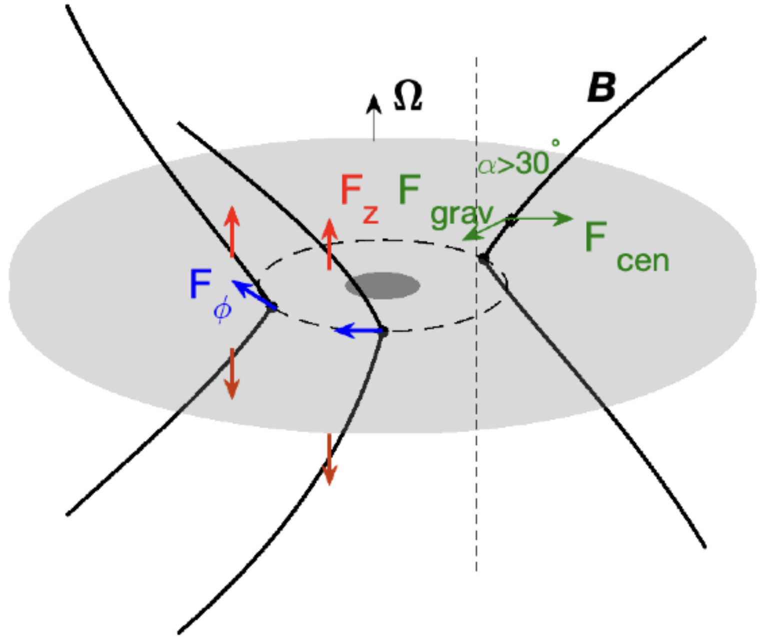
https://github.com/PinghuiHuang/athena-multifluid-dust

Ambipolar Diffusion (AD) controls the non-ideal MHD in outer disks.

MHD Disk Wind



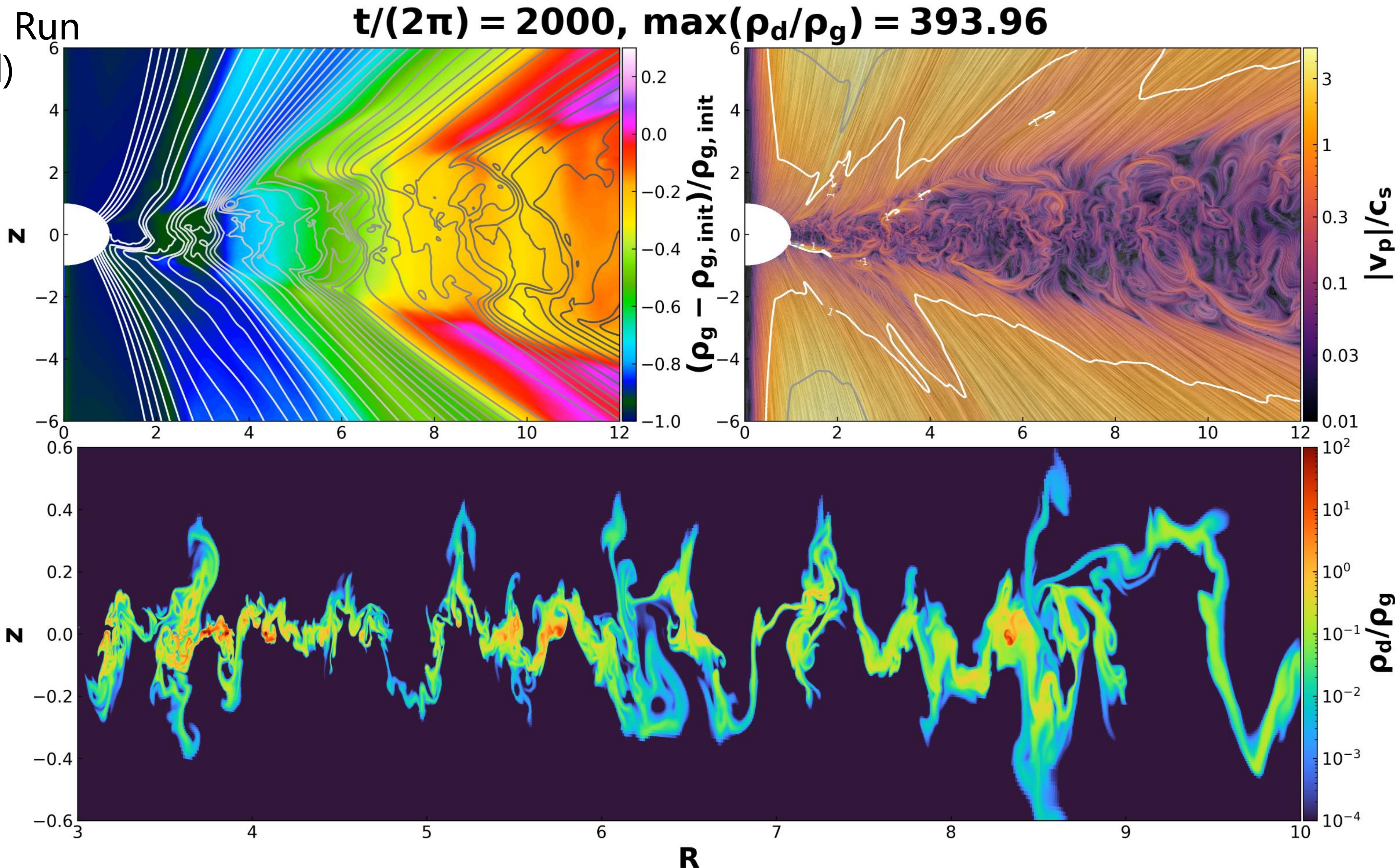
Requests a large-scale net poloidal field threading the disk.

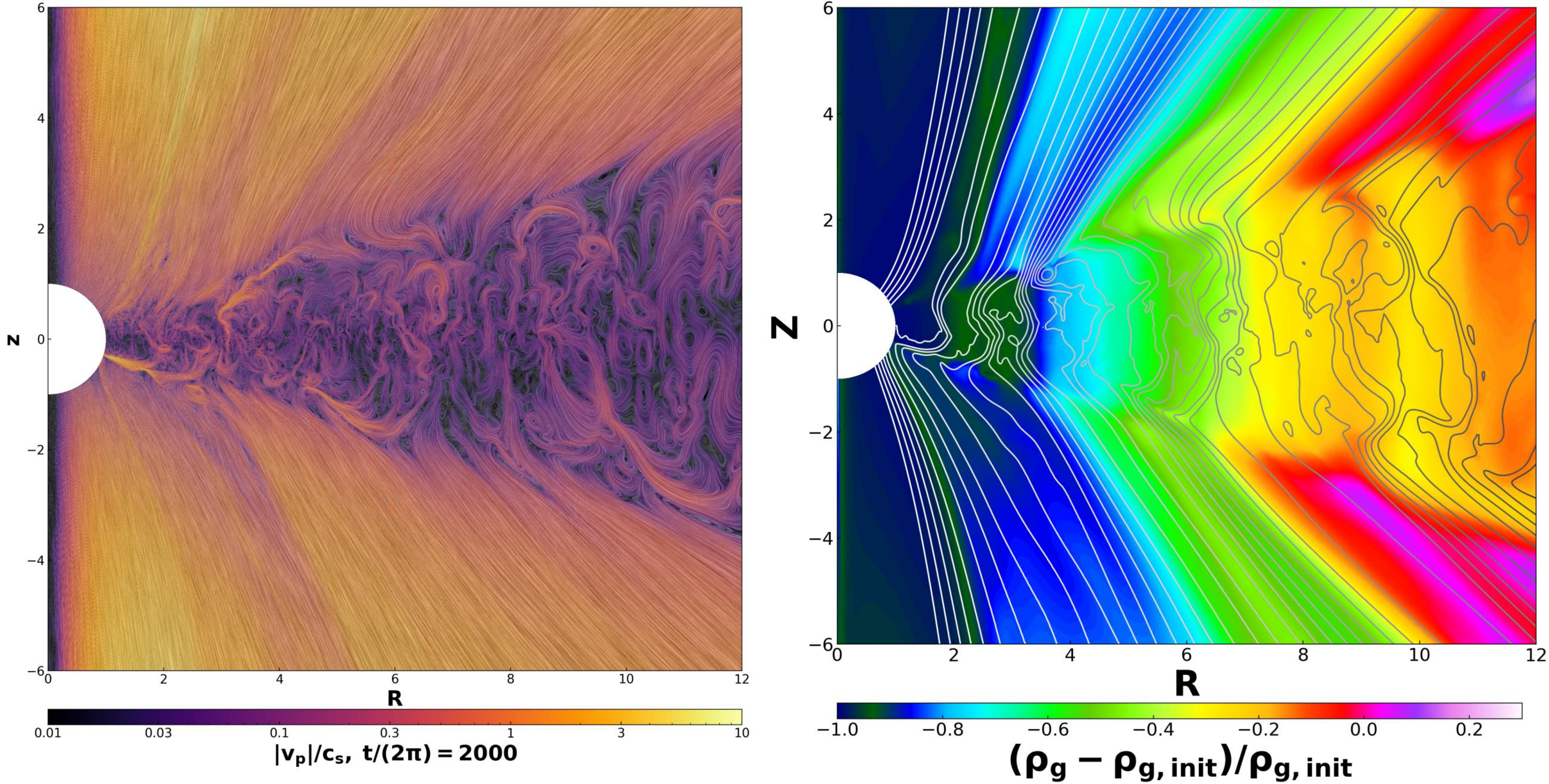


Bai, X. ARAA, submitted.

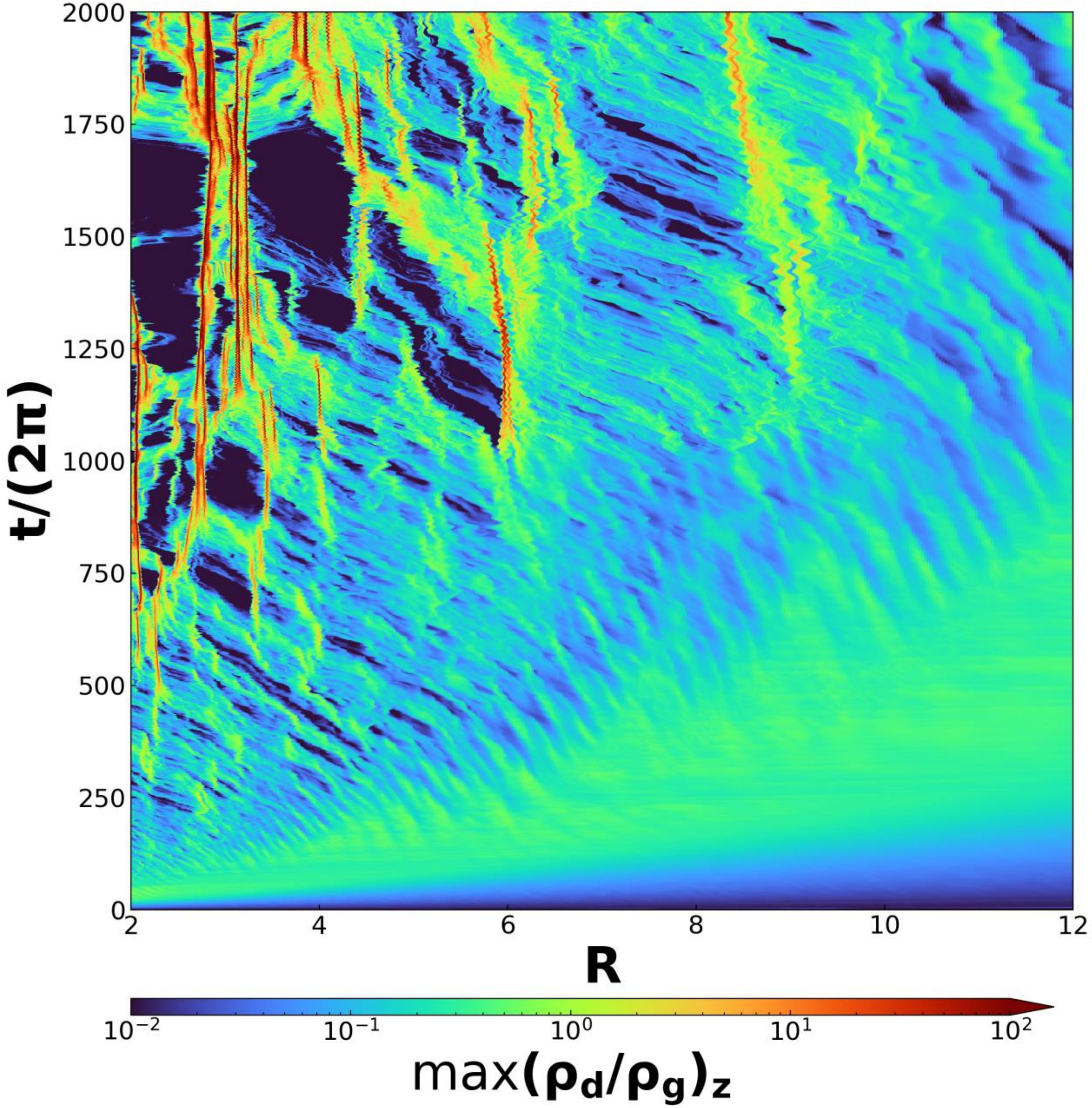
The MHD disk wind can exert a vertical torque that extracts angular momentum from the disk, which is more efficient than the turbulent viscous torque.

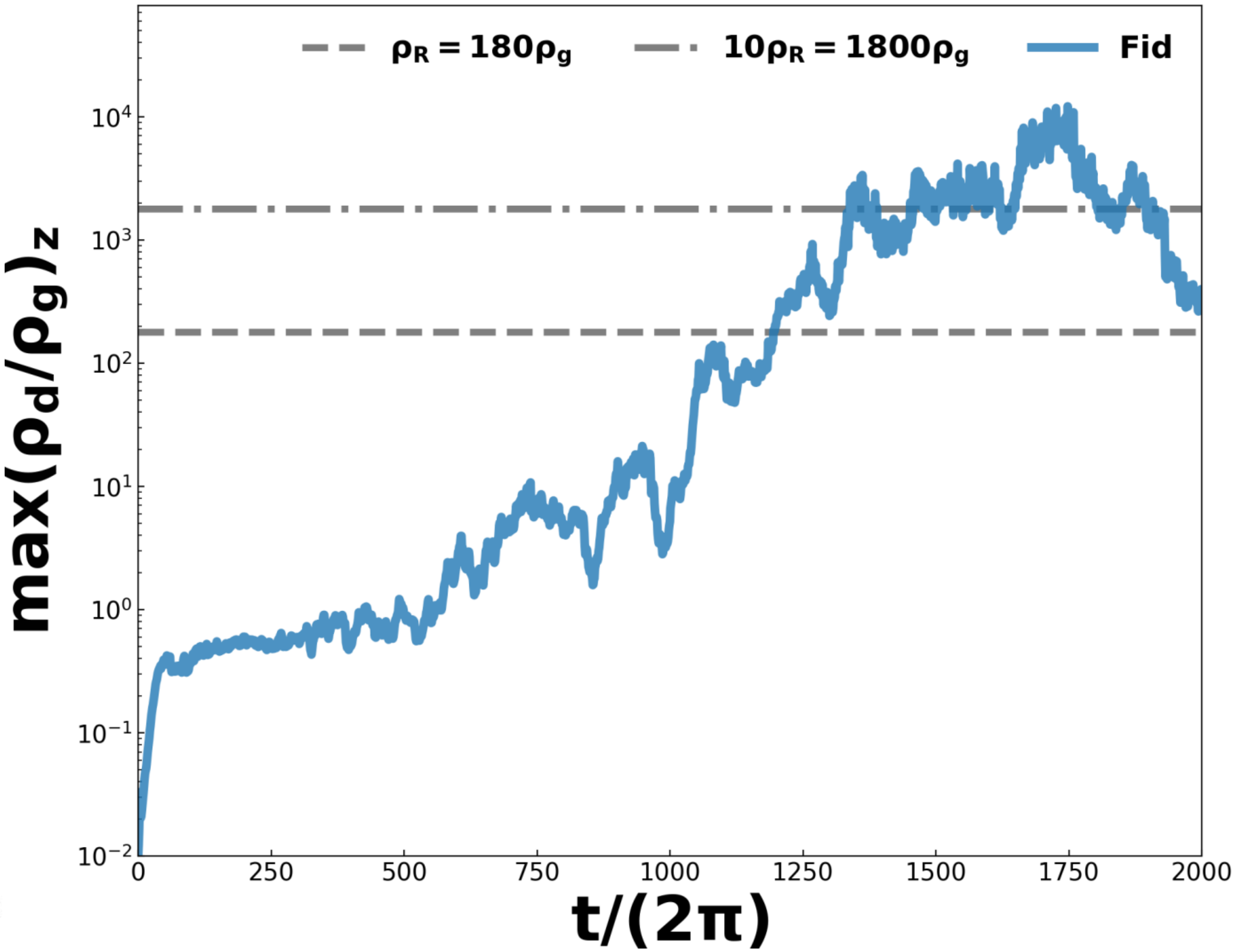
Fiducial Run (Fid)





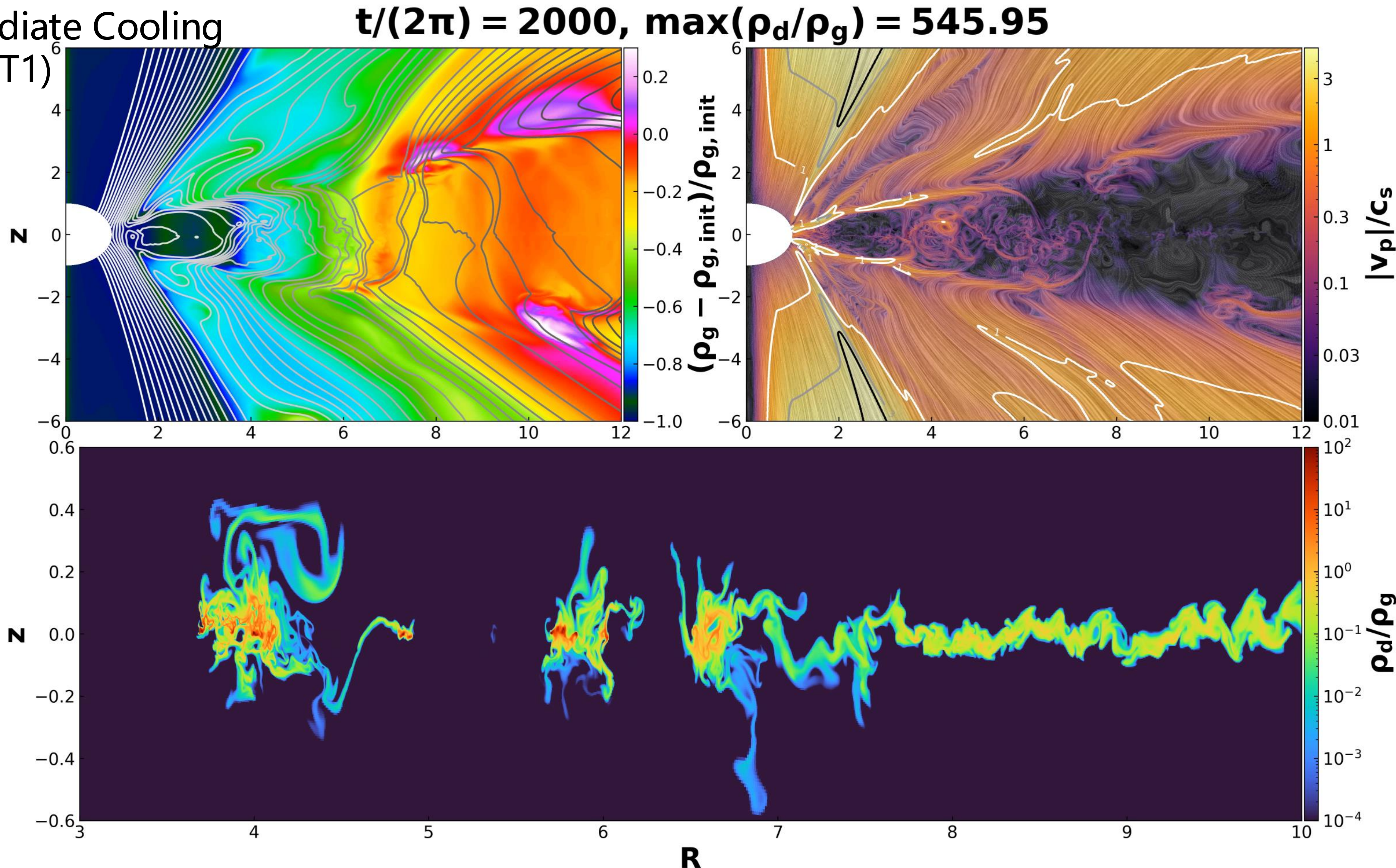
The disk is turbulent and windy, characterized by weak VSI activity and mild magnetic flux concentration (MFC).

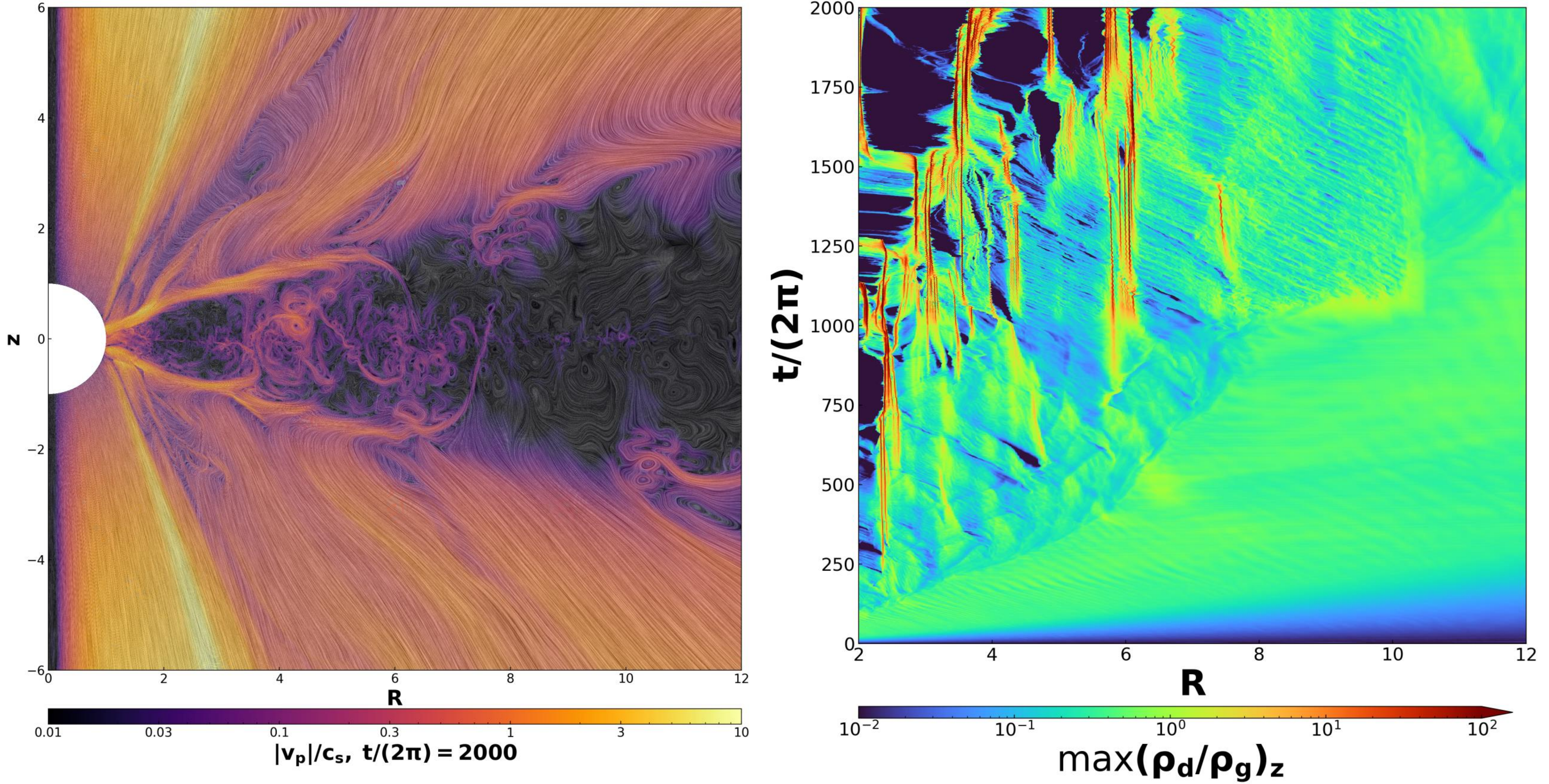




Left and Right: Space–time (R–t) diagram and the time evolution of $max(\rho_d/\rho_g)$.
The maxima value of ρ_d/ρ_g could easily reach the critical Roche density (Li & Youdin 2021).

Intermediate Cooling
$\beta_{th} = 1$ (T1)





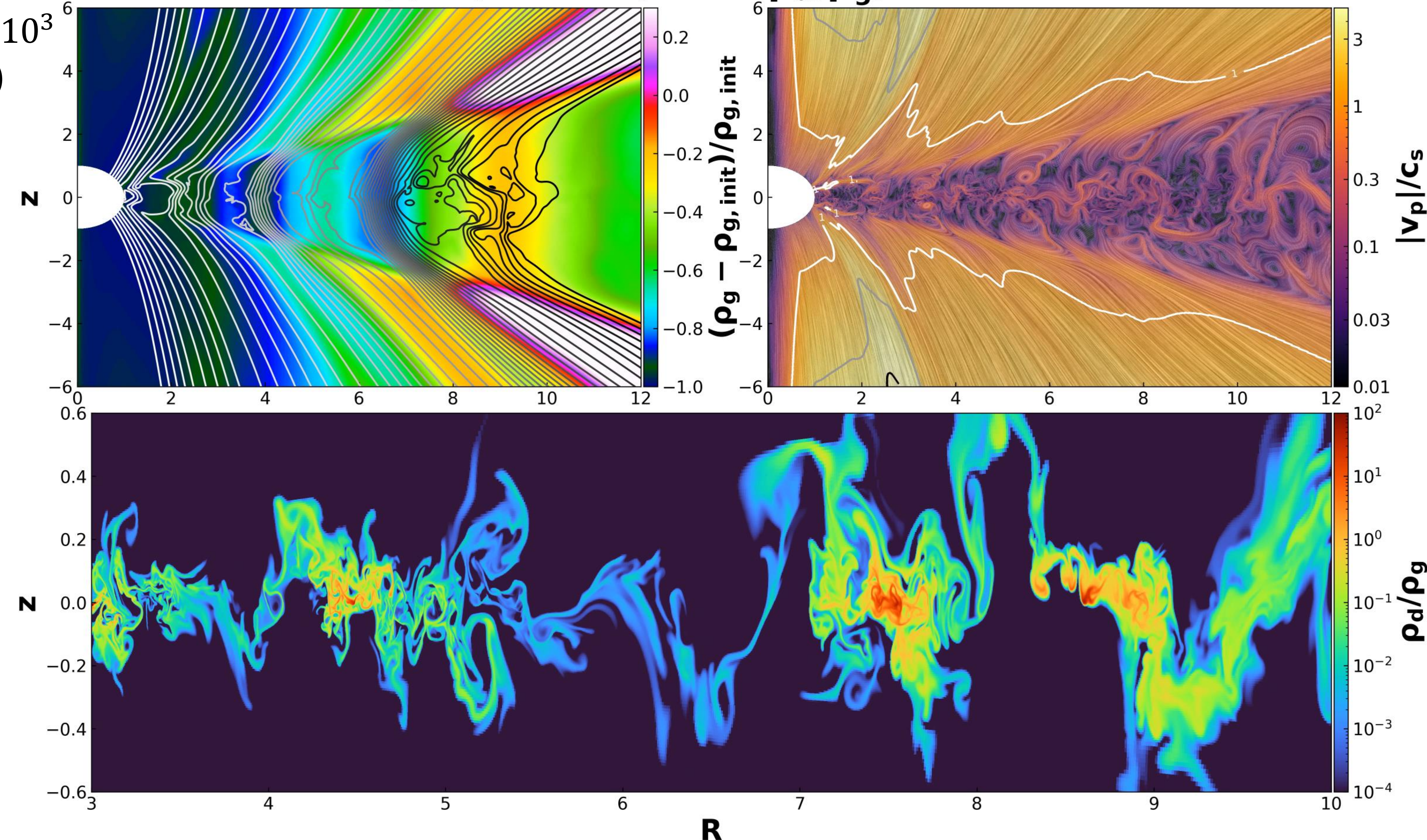
The disk is more laminar. The VSI dies out, and the interaction between the SI and MFC leads to strong dust clumping.

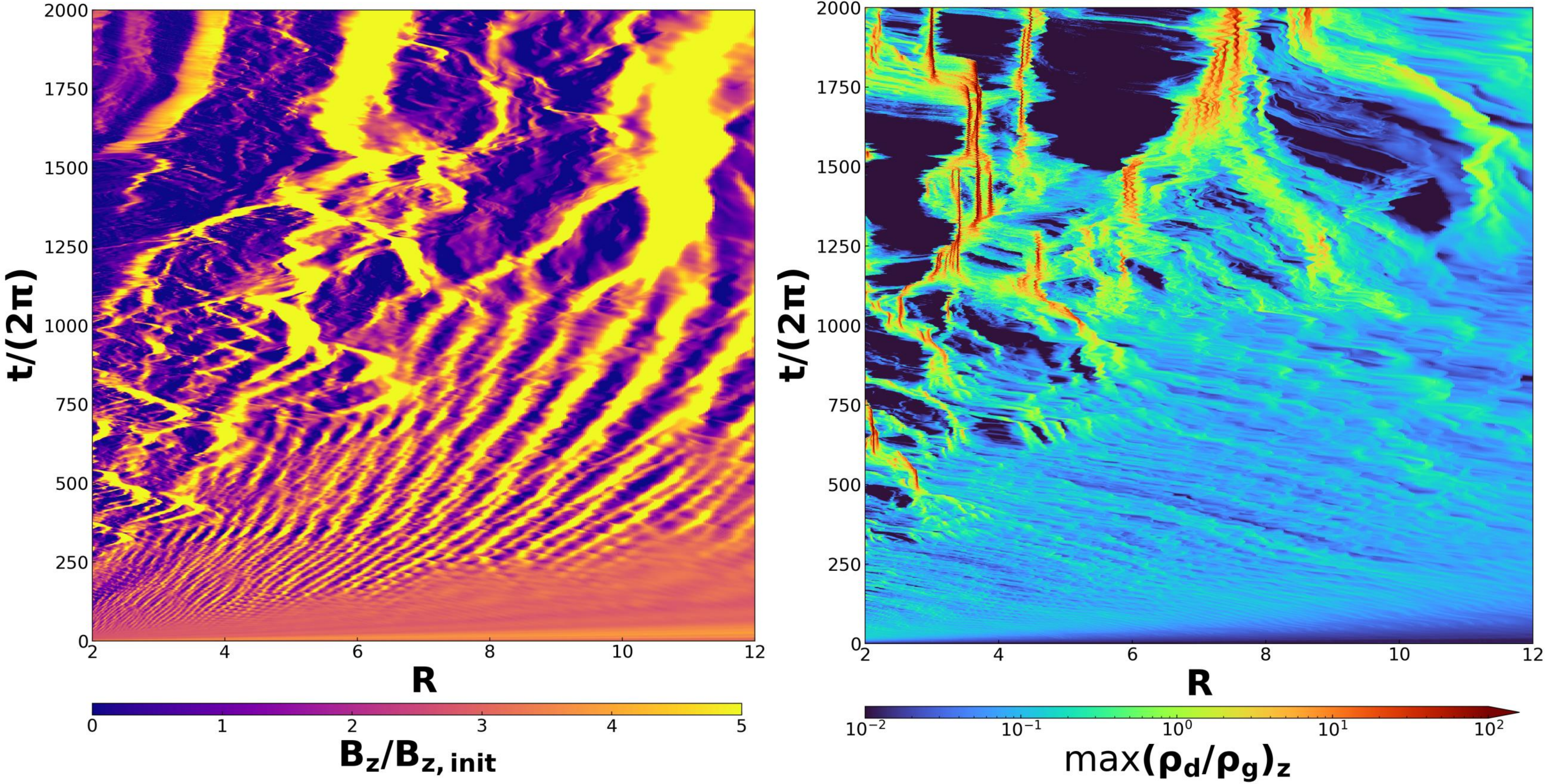
Strong B Field

$\beta_{pl} = 10^3$

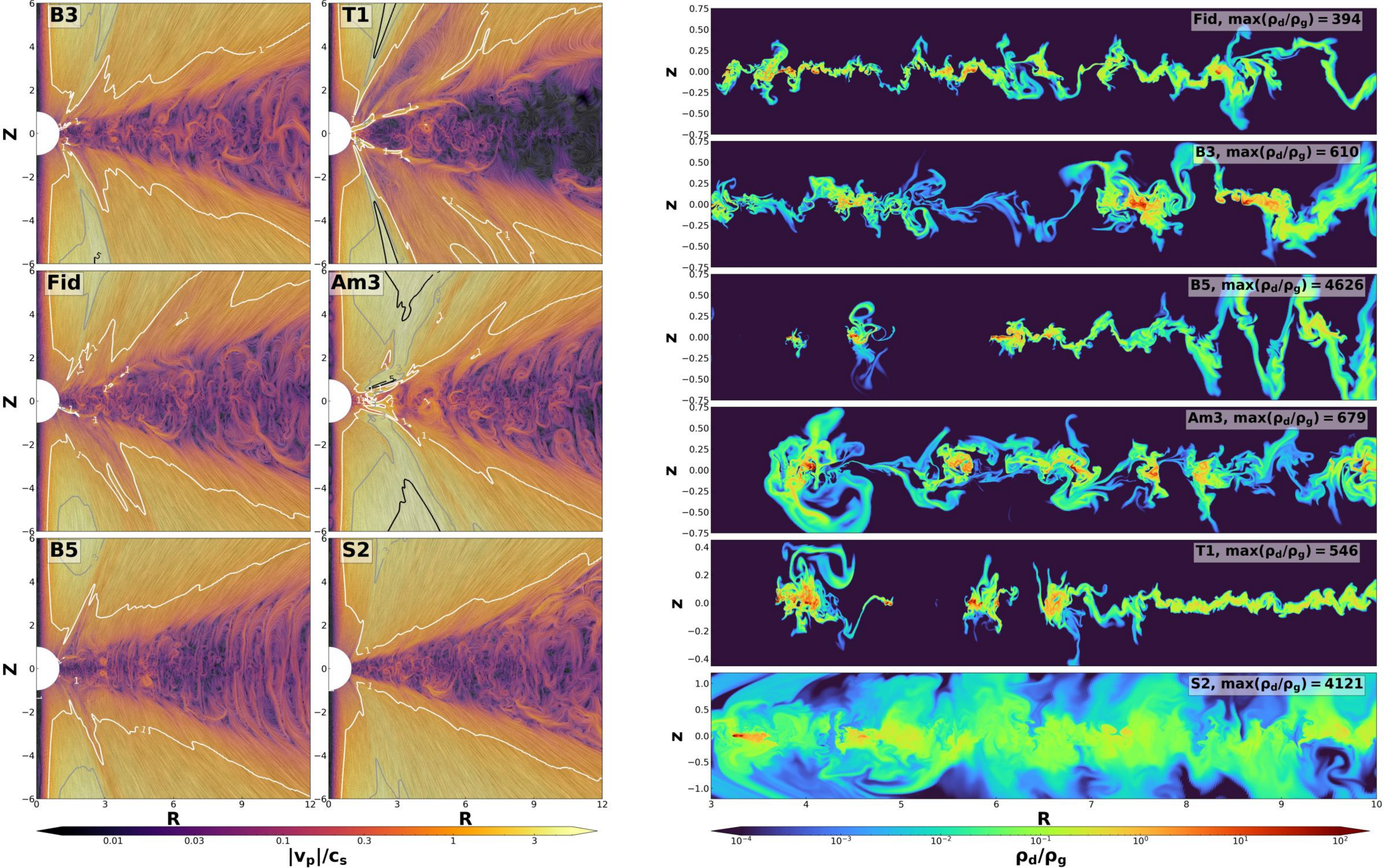
(B3)

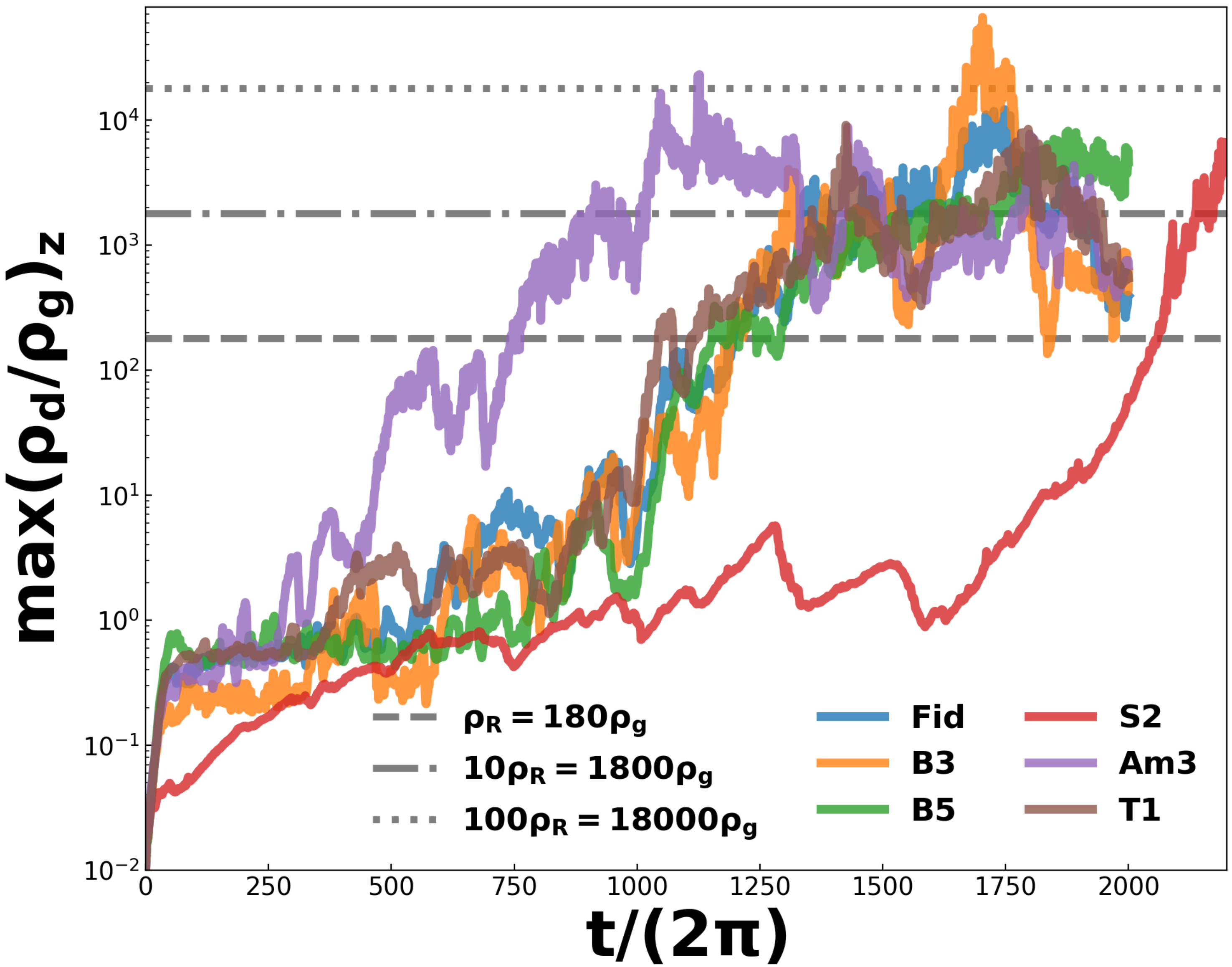
$t/(2\pi) = 2000$, $\max(\rho_d/\rho_g) = 610.33$





Significant MFC occurs within gas gaps.
Dust is trapped in the zonal flows, leading to strong dust concentration.


B3
T1
Fid
Am3
B5
S2
Z
R
$|v_p|/c_s$
0.01
0.03
0.1
0.3
1
3
Fid, $\max(\rho_d/\rho_g) = 394$
B3, $\max(\rho_d/\rho_g) = 610$
B5, $\max(\rho_d/\rho_g) = 4626$
Am3, $\max(\rho_d/\rho_g) = 679$
T1, $\max(\rho_d/\rho_g) = 546$
S2, $\max(\rho_d/\rho_g) = 4121$
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
ρ_d/ρ_g



Dust Concentration and Planetesimal formation could be more efficient under the realistic gas dynamics!

Conclusions:

- Understanding **realistic gas dynamics** is crucial for studying **planetesimal formation** in protoplanetary disks.

- MHD disk turbulence/wind with AD **reduce gas density** and **promote dust concentration** in all cases, **favoring planetesimal formation**.

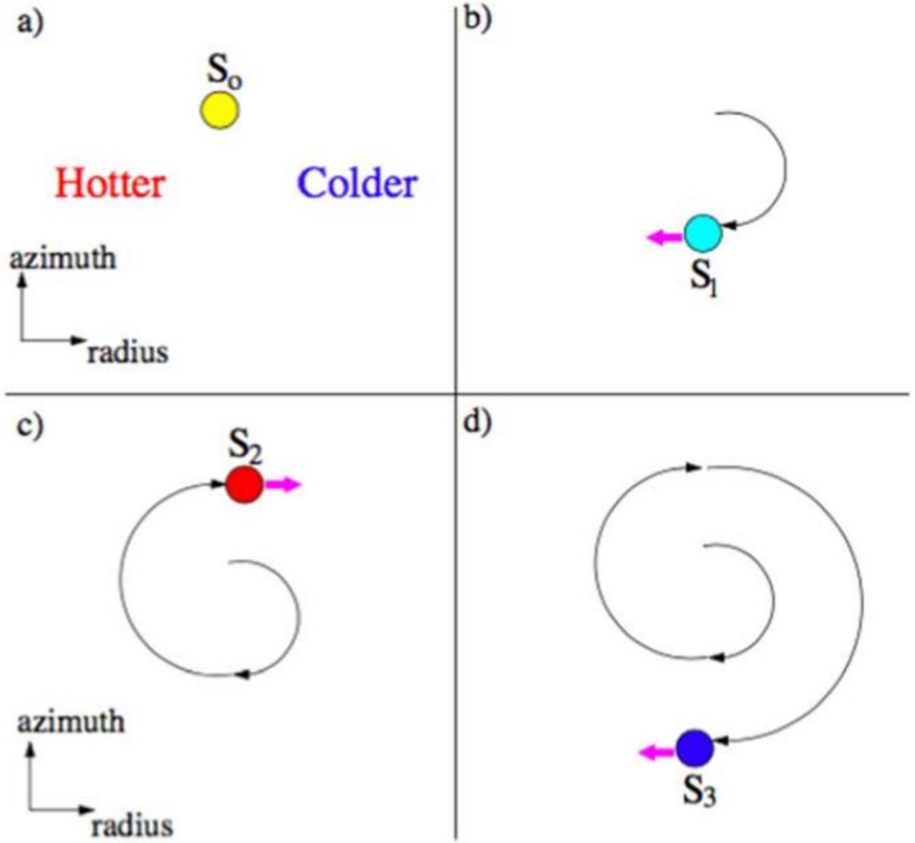
- In weakly magnetized disks, the **SI operates with MHD wind**, **with or without VSI**, **produces strong dust clumping**.

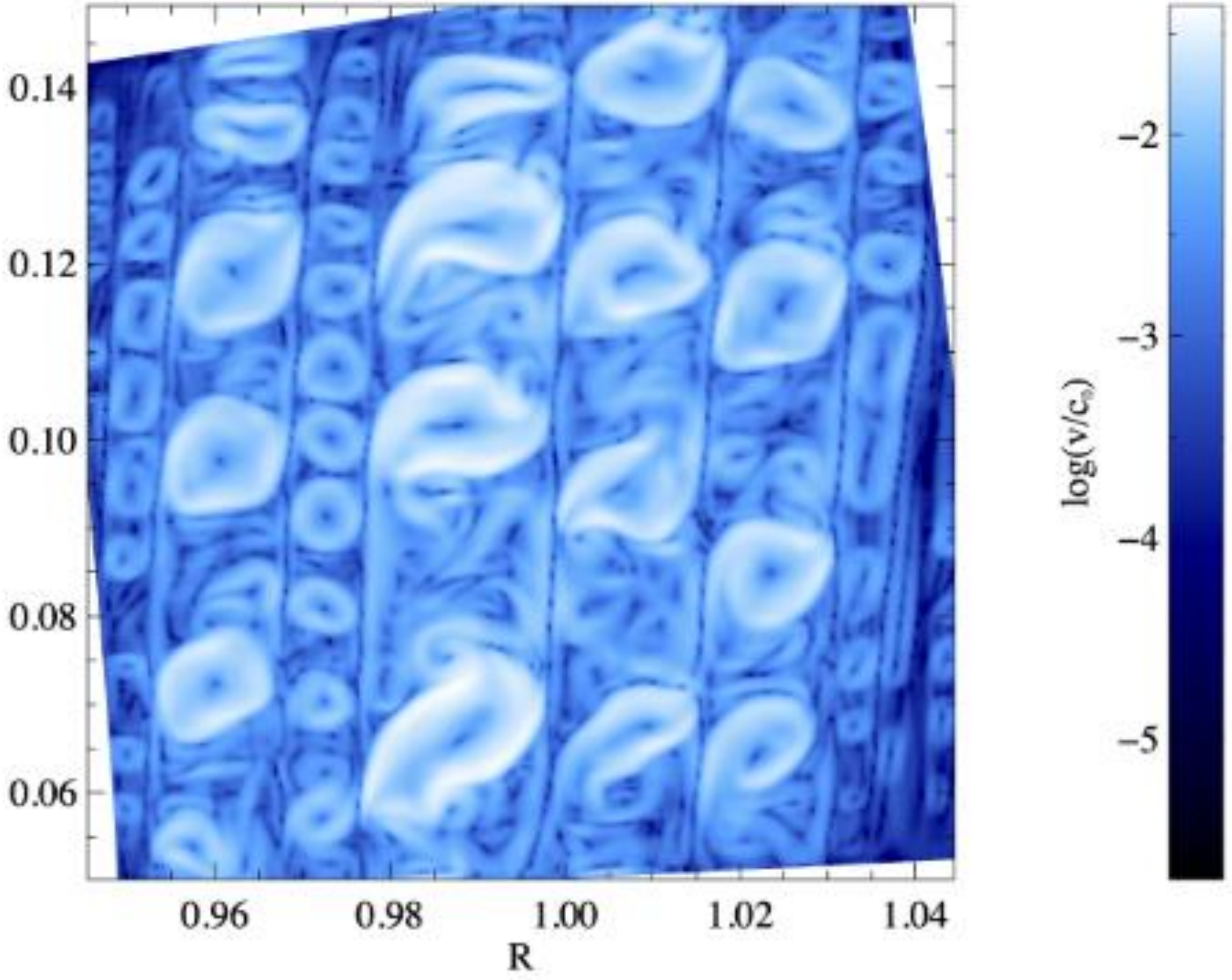
- In strongly magnetized cases, the **VSI is damped**, while **MFC gives rise to zonal flows**. These flows significantly **reduce or halt radial dust drift and weaken or suppress the SI**; however, **dust concentration** within the zonal flows may still **reach densities sufficient to initiate planetesimal formation**.

Next:

3D effects (MRI dynamo, RWI vortices, KHI destruction), Ohmic & Hall

Convective Overstability (COS)





Similar to Rayleigh-Bénard Convection

Negative Entropy Gradient
Maximal growth rate happens at $\gamma\beta_{th} \sim 1$ (Lyra & Umurhan 2019)

A global negative entropy gradient is not necessary for COS in stratified disks (Klahr et al. 2026 a,b).

III. Dust Clumping in Thermally Buoyant Disks

COS+GSF+SI

Huang & Bai, in prep.